

1. VALUE ANALYSIS

What is it?

- One of the **major techniques of cost reduction**
- A disciplined approach that ensures **necessary functions at minimum cost** without compromising on **quality, reliability, performance, and appearance**
- Also known as: **Value Engineering, Value Assurance, Value Management**

Core Idea (Flowchart)

Value Analysis → Function of Product/Services → Required Function → VALUE
→ Lowest Overall Cost → VALUE

What is "Value"?

Remember: P-R-A-S-S-R

- Performance
- Reliability
- Appearance
- Service Life
- Safety
- Range of Operation

Aims of Value Analysis (5 points)

1. **Simplify** the product
2. Use **cheaper and better** materials
3. **Improve** product design
4. **Reduce** the product cost
5. **Increase** the profits

When to Apply Value Analysis? (5 triggers)

1. Company products are **losing market** / decline in sales
2. Products are **priced higher** than competitors
3. **Manufacturing cost is high**
4. **Decreasing profitability**
5. Company is **unable to meet delivery schedules**

Uses of Value Analysis (5 benefits)

1. **Reduces** cost of product
2. **Balances** cost and performance
3. **Motivates employees** to come out with creative ideas
4. **Helps satisfy customers** with company's product
5. **Prevents over-design** of components

2. BUSINESS ORGANIZATIONS

Definition

- A setup to perform specific functions to **generate profit** and provide **service**
- Functions under **relevant laws** (Companies Act 2017)
- Must register with statutory body (e.g., PEC for civil engineering)

Entrepreneurship

- **Entrepreneur** = person who takes **risk** to start business for **profit + service**
- **Entrepreneurship** = process to **plan, design, launch, manage** a business

4 Types of Entrepreneurships

Type	Key Feature
Small Business	Privately owned, fewer employees, less income
Scalable Medium	Can grow without being hurt by increased demand
Large Business	500+ employees, manufacturing/industries
Social Entrepreneurs	Community/social service (e.g., Grameen Bank, SSUET, SKMH)

4 Types of Business Organizations

A) Sole Proprietorship (1 owner) *Simplest form — ONE person does everything*

Merits:

1. Easy to start & close
2. Quick decisions
3. Small capital needed
4. Close customer contact
5. Business secrets stay safe
6. No labor disputes

Demerits:

1. No division of labor
2. No large-scale advantages
3. High production cost
4. Limited machine use
5. Less credit worthiness
6. Uncertain continuity

B) Partnership (2+ persons) *Share profits AND losses in agreed proportion*

Merits:

1. Easy to set up
2. More capital available
3. Large production possible
4. Use of diverse talents
5. Promptness + checks on hasty decisions
6. Personal interest and initiative

Demerits:

1. Lack of permanency
2. Lack of mutual confidence
3. Unlimited liability
4. Delay in decisions
5. Difficult to close down

C) Private Limited Company (2–50 shareholders)

- Liability **limited** to extent of shares
- **Cannot** raise funds from stock exchange
- Privately held small business entity

D) Public Limited Company / Joint Stock Company (unlimited shareholders)

- Shares offered to **general public** via stock market
- Listed on stock exchange (e.g., KESC, PTCL)
- **Can** raise funds through shares and bonds
- Needs minimum **7 promoters**
- Must submit **annual report** to Registrar of Companies
- Also called **corporate form** of business

Merits:

1. Can mobilize large resources
2. More permanency
3. Specialized management

4. Easy to collect capital
5. Transferable shares
6. Suitable for large enterprise

Demerits:

1. Ownership vs management gap
2. Shareholders lack of interest
3. Exploitation of majority shareholders
4. Heavy taxes
5. Encourages speculation
6. Corruption & labor disputes possible

3. THEORY OF DEPRECIATION & DEPLETION

Key Definitions

Term	Meaning
Asset	Anything owned from business point of view
Tangible Asset	Physical existence + value (machinery, building, pipes)
Intangible Asset	No physical form but has value (trademark, goodwill, skill, knowledge)
Natural Asset	Land, water, air, minerals, mines

Depreciation vs Depletion

	Depreciation	Depletion
What	Decrease in worth of **property/asset**	Decrease in worth of **natural asset**
Cause	Wear & tear, deterioration, obsolescence	Extraction/use of natural resources
Example	Machine losing value over years	Water reducing in dam, mine running out

Purpose of Depreciation/Depletion

1. To **recover capital** that was invested
2. To work out the **real cost** of assets

Two Types of Depreciation

Type	Meaning	Cause
Physical	Loss of **physical ability** to perform	**Wear & tear** — related to time & maintenance
Functional	Loss of **demand** for the asset	**Technology advancement** (e.g., Pentium 2 → Pentium 5)

Important Values — BV, MV, SV

Value	What it means	Key point
Book Value (BV)	Original cost – total depreciation charged	Current worth **on company records** ; decreases yearly
Market Value (MV)	Price if sold on **open market**	Can differ from BV; land **appreciates** while BV depreciates
Salvage Value (SV)	Net residual value after asset is **fully used**	Sale price at end of useful life

3 Methods to Calculate Depreciation

Method	Key Feature
Straight Line (SLD)	Constant amount each year; BV reduces **linearly**
Declining Balance (DB)	Higher depreciation in **early years**
Sum of Year Digits (SYD)	Rapid write-off; most value gone in **first 1/3 of life**

SLD Key Idea

- Same depreciation **every year**
- $D = (\text{Initial Cost} - \text{Salvage Value}) / \text{Life of Asset}$
- Book Value after m years = Initial Cost – (m × D)

SYD Key Idea

- **Larger** depreciation in early years, **smaller** in later years
- Most of the asset value written off in the **first 1/3rd** of its life
- Uses sum of digits (1+2+3+...+n) as denominator

4. CAPITAL FINANCING & ALLOCATION

Definition

- Process of **allocation and distribution of financial resources** for **long-term financial stability**
- Purpose: **value addition** + **fair returns** to capital providers while covering risks

Key Points

- Capital decisions → **higher management level** and **country policy level**
- Investment decisions are **strategic/policy decisions**
- Invest where yield is **positive** or better than **MARR** (Minimum Attractive Rate of Return)
- Always **anticipate risks** before investing — what can affect **cash inflow/outflow**?

3 Modes of Financing

Mode	How it works
Self-investment	Investor uses own money
Shareholders	Equity from multiple investors
Borrowing	Bank/financial institution — lease/credit with compound interest

Budget/Financial Control

- Ensures **financial discipline** for reliable capital financing

Benefit Cost Ratio (B/C or BCR)

- Used to **evaluate projects** — are benefits greater than costs?
- **B/C > 1** → Project is **DESIRABLE** ✓
- **B/C < 1** → Project is **NOT desirable** ✗

Method	Formula	Note
Conventional	$B/C = B \div (CR + O\&M)$	Older method; cannot calculate dis-benefits
Modified	$B/C = (B - DB - O\&M) \div CR$	Newer; accounts for **dis-benefits (DB)**

CR = Capital Recovery Cost (initial investment converted to annual cost) O&M = Operation & Maintenance expenses

5. GOODS MARKET & MONEY MARKET

Goods Market

- All **buying and selling** of goods & services
- Households = **buyers**, Companies = **sellers**
- Determines **equilibrium level of output**

Money Market

- Interaction between **demand for money** and **supply of money**
- Supply controlled by **State Bank** through banking system
- Determines **equilibrium interest rate**
- Provides **short-term liquidity** to governments, banks, organizations
- Securities trade in **very high denominations**
- It's a **dealer/trader market** — firms trade at their own risk

Key Difference

	Goods Market	Money Market
Determines	Equilibrium **output**	Equilibrium **interest rate**
Players	Households & companies	Banks, government, large organizations
What's traded	Goods & services	Money/securities

6. MARKET STRUCTURE (4 Types)

Memory trick: P-M-O-M (Perfect, Monopoly, Oligopoly, Monopolistic)

Structure	# of Firms	Key Features
Perfect Competition	100s of firms	Free entry/exit, homogeneous goods, perfect information, **normal profit**
Monopoly	1 firm ($\geq 25\%$ market share)	Barriers to entry, higher prices, **economies of scale**
Oligopoly	Few large firms (5-firm ratio $> 50\%$)	Firms are **interdependent** , some barriers to entry
Monopolistic Competition	Several firms with brand loyalty	Low barriers, **differentiated products** , less profit than monopoly